

Simple Path Structural Encoding for Graph Transformers

Count paths, not walks

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Reading companion · GNN / Graph-Transformer thesis exploration

★ TL;DR

Graph transformers inject structure into attention through an **edge encoding**. The incumbent is **RWSE** — stack the k -hop random-walk matrices $P_k = (D^{-1}A)^k$ and feed them to the attention layer. RWSE is cheap (closed form) but **ambiguous**: the authors prove that every node pair in an even-length **cycle** has an RWSE-equivalent pair in a **path graph**, so walk probabilities cannot separate cycles from chains. **SPSE** replaces P_k with S_k , the count of **simple paths** (no repeated nodes) of length k between i and j . For adjacent pairs this count **is** cycle counting (Prop. 3), which is exactly the structure molecules are made of. Simple-path counting is $\#P$ -hard in general, so they give an **approximate** counter: decompose the graph into many DAGs by mixed DFS/BFS traversals, count by matrix powers on each DAG, keep the running **maximum**. Drop-in replacement, **zero extra parameters**: improves 21 / 24 benchmark cases, significantly on molecular data.

♦ The big question

Graph transformers see all pairs, so **distance** is not the problem — **which structure** a pair sits inside is. Can an edge encoding be made to carry local **cyclic** structure that random walks provably blur away?

♦ The one-line answer

Yes — count **simple paths** instead of walks. Path counts between adjacent nodes are cycle counts, and an approximate DAG-decomposition counter makes them affordable enough to precompute once.

1 · Setup — where the encoding actually enters

► Definition — Walk vs. simple path

A **walk** of length k is any node sequence $v_0 \dots v_k$ with consecutive nodes adjacent — **revisits allowed**. A **simple path** is a walk in which **all nodes are distinct**. A **cycle** is a walk whose nodes are distinct except $v_0 = v_k$.

► Definition — The two encoding matrices

Random walk matrix $P_k = (D^{-1}A)^k \in \mathbb{R}^{|V| \times |V|}$ — landing **probabilities** of length- k walks. Closed form, cheap.

Simple path matrix $S_k \in \mathbb{N}^{|V| \times |V|}$ — $(S_k)_{ij}$ is the **number of simple paths** of length k from i to j . No closed form; computing it is expensive (Vassilevska & Williams 2009).

Both are stacked over $k = 1 \dots K$ and pushed through a small network ϕ_0 to give an encoding matrix $E \in \mathbb{R}^{|V| \times |V| \times d}$, which enters **both** the attention logits and the values:

$$\begin{aligned}a_{ij} &= \phi_1(W^Q x_i, W^K x_j, (E)_{ij}) \\ \alpha_{ij} &= a_{ij} / \sum_k a_{ik} \\ y_i &= \sum_j \alpha_{ij} \phi_2(W^V x_j, (E)_{ij})\end{aligned}$$

📌 My note

The encoding is **not** just an attention bias — it is mixed into the **value** path too (eq. 3). That is what makes this a genuine “structural edge encoding” rather than a distance prior, and it is why SPSE can be a **drop-in**: swap $E_{RW} \rightarrow E_{SP}$ in equations 1 and 3, change nothing else. Parameter count is **identical**.

💡 Thesis hook

Our `GlobalAttnConv` sits at the **poorest** end of this family. We add a learned per-head bias over **shortest-path-distance buckets** (`spd_bias`, `src/layers.py:179`, added pre-softmax at :196) — a single scalar per (bucket, head), logits only, never touching values. The ladder is:

SPD bucket bias $\rightarrow$ **RWSE** $\rightarrow$ **SPSE**

SPD keeps **one number** (the distance); RWSE keeps a K -vector of walk probabilities; SPSE keeps a K -vector of path counts. Each step up strictly refines the previous. Our SPD bias flattens everything the paper cares about: two pairs at distance 2 get the same bias whether they sit in a triangle-rich blob or a tree.

2 · The limitation — RWSE cannot see a cycle

The core negative result. Random walks **normalise**: P_k holds probabilities, not counts, so the mass leaving a node is always 1 and structural differences can cancel.

◆ Proposition — 1 (even cycle = path)

Let G be an **even-length cycle** on $|V| = 2n$ nodes and G' a **path graph** on $2n + 1$ nodes. Then for **any** pair (i, j) in G there is a pair (i', j') in G' with $(i, j) \stackrel{RW}{=} (i', j')$ — i.e. identical P_k entries for **all** k .



Prop. 1: the marked pairs carry **identical** random-walk encodings at every k , though one lies on a cycle and the other on a chain. SPSE separates them immediately — the cycle pair has two simple paths between its endpoints, the path pair has one.

◆ Proposition — 2 (general ambiguity)

For **any** graph G and pair (i, j) there exists a **non-isomorphic** G' and a pair (i', j') with $(i, j) \stackrel{RW}{=} (i', j')$. So an RWSE edge encoding **never** identifies a graph.

📌 My note

Prop. 2 is weaker than it first sounds — the authors immediately concede it is “also obviously true for SPSE”. No pairwise encoding identifies a graph. The **load-bearing** result is Prop. 1, which is a **concrete, structured** collapse (cycles $\rightarrow$ chains), not a generic pigeonhole. Read Prop. 2 as framing, Prop. 1 as the actual gap.

3 · Why path counts are cycle counts

◆ Proposition — 3 (path counts on an edge = cycle counts)

Let $(i, j) \in E$ be an **adjacent** pair and $(S_k)_{ij} = m_k$. Then for $k \geq 2$ there are **exactly** m_k cycles of length $k + 1$ in G that use (i, j) as an edge. ($k = 1$ just counts parallel edges.)

This is the paper’s real engine. For an edge, “how many simple paths of length k join my endpoints” and “how many $(k + 1)$ -cycles do I sit on” are **the same number**. So SPSE hands the attention layer a per-edge **cycle spectrum** for free.

✍ My note

Crucially Prop. 3 holds **only for adjacent pairs** — $(i, j) \in E$. For non-adjacent pairs the path counts are still informative but no longer a clean cycle count. And no RWSE analogue exists: landing probabilities on a cycle can be driven arbitrarily low just by adding nodes to it, so the signal is not scale-stable.

💡 Thesis hook

This is why the paper’s benefit concentrates on **molecules** (ZINC, Peptides, PCQM4Mv2): chemistry is cycle structure — aromatic rings, functional groups. Their Fig. 2 shows SPSE giving identical encodings to the two C=O bonds of two carboxylic acids, and to edges in two different 6-atom rings. That is chemically correct invariance, learned for free from the encoding rather than from data.

4 · Making it computable — DAG decomposition

Exact simple-path counting is intractable (the count itself can grow like $(|V| - 2)! / (|V| - k - 1)!$). The workaround exploits one fact: **on a DAG, path counts between all pairs are just powers of the (directed) adjacency matrix**, $O(K |V|^3)$.

So: turn the undirected graph into **many** DAGs, count on each, and keep the best.

► Definition — DAGDECOMPOSE (Alg. 2)

From a root r , build a node ordering π by **mixing traversals**: run $d_{\text{DFS}} - 1$ steps of **DFS**, then a **partial BFS** step (expand only a random subset of children), then **BFS** as far as possible; repeat until all nodes are visited. Orienting every edge toward the higher index yields a DAG. Sweep $d_{\text{DFS}} = 0 \dots D_{\text{DFS}}$, repeat N times per depth (random effects), over $R \cdot |V|$ sampled roots.

Why the mixture: **DFS** discovers long paths but commits to one branch and so tends to find only **one** path between a pair; **BFS** explores breadth-first but **misses long paths**. The **partial BFS** step is the trick that makes otherwise unreachable paths traversable (their Fig. 3, on a Circular Skip Link graph).



running **max** over DAGs $\rightarrow$ lower bound on true count

Cost. $O(KRD_{\text{DFS}}N |V|^3)$. The $RD_{\text{DFS}}N$ factor is “tens to hundreds”, so SPSE is **much** more expensive than RWSE — but it is a **one-time preprocessing step**, and vastly cheaper than the $2^{|E|}$ possible decompositions. Wall-clock from Table 2: **1 h** for ZINC, **80 h** for CIFAR10 and PCQM4Mv2 (3.7M graphs).

Encoding the counts (eq. 4)

Counts explode, so they are squashed by **iterated** logarithms before hitting the network:

$$f : x \mapsto \alpha g^n(x) + \beta, \quad g : x \mapsto \ln(1 + x)$$

with g^n meaning g composed n times. Per Table 2: denser datasets want **higher n** (more squashing) — $n = 1$ for ZINC/PCQM4Mv2, $n = 3$ for PATTERN/CLUSTER/MNIST/CIFAR10.

✍ My note

n is doing real work: it is the knob that keeps a count-based encoding from being dominated by hub pairs in dense graphs. Compare our own preprocessing — we normalise nothing on the SPD buckets because a distance is already $O(\text{diam})$. The moment we move to counts, this compression question becomes ours too.

5 · Results

Synthetic — cycle counting (validates Prop. 3)

12,000 graphs (avg. 149 nodes / 190 edges), built by injecting cycles of length 3–8 until each length’s count lands in $[0, 14]$; six simultaneous multiclass tasks (“how many k -cycles?”). SPSE beats RWSE in **all but one** configuration, on both GRIT and CSA.

✍ My note

The one revealing detail: at the **largest** model config, all models get “almost perfectly”, so the gap narrows. Their reading — **deep architectures can compensate for expressivity limits in the encoding**. That is the same depth-vs-prior trade we keep hitting: a weaker encoding is survivable if you pay for it in layers.

Real benchmarks

Eight datasets: molecular (ZINC, Peptides-func, Peptides-struct, PCQM4Mv2), SBM (PATTERN, CLUSTER), superpixel (MNIST, CIFAR10). SPSE swapped into **GRIT**, **CSA**, and (partially) **GraphGPS**. Retrained on **10 seeds, no hyperparameter tuning**.

Benchmark	RWSE	SPSE	note
ZINC (GRIT, MAE ↓)	0.065 ± 0.005	0.059 ± 0.001	significant
ZINC (CSA, MAE ↓)	0.069 ± 0.003	0.061 ± 0.003	significant
Peptides-func (GRIT, AP ↑)	0.6803 ± 0.0085	0.6945 ± 0.0113	significant
Peptides-struct (GRIT, MAE ↓)	0.2486 ± 0.0012	0.2449 ± 0.0018	significant
CIFAR10 (GRIT, acc ↑)	76.246 ± 0.954	77.022 ± 0.430	significant
PATTERN (CSA, acc ↑)	87.008 ± 0.062	87.064 ± 0.052	significant
CLUSTER (GRIT, acc ↑)	79.730 ± 0.189	79.571 ± 0.122	no gain — dense

Headline: better in 21 of 24 cases, statistically significant in 5 of 6 molecular cases for CSA and GRIT. Gains are largest exactly where cycles carry meaning.

✍ My note

Two honest asterisks the authors raise themselves:

GPS gains are muted in GraphGPS the encoding is restricted to E and used only for **node-level** PE in the MPNN layer. Since parameters are held fixed, adding SPSE there effectively **shrinks** the existing edge embedding dimension. The lesson they draw — and it matters for us — is that **feeding the full encoding matrix into self-attention beats restricting it to real edges**.

CLUSTER gets nothing densely connected SBM graphs defeat the approximate counter.

6 · Limitations — where the approximation bites

The counter keeps a running **maximum** over DAGs, so the reported counts are **lower bounds** on the true counts. Worse, some path sets are **unreachable by any single DAG**: their Fig. 6 shows two length-4 paths

between a pair whose simultaneous discovery would require a **cyclic** orientation — the algorithm can only count them one at a time.

? Open question

Density is the failure axis. In dense graphs (CLUSTER, PATTERN) many paths are missed, counts are underestimated, and the encoding degrades toward noise. Their own suggestion: **trade path length for accuracy** — shorter K with exact counts may beat longer K with bad estimates. Untested by them. That is a cheap, well-posed experiment.

? Open question

Appendix B is refreshingly candid: a Graziani-style argument shows global attention with SPSE is more expressive than 1-WL — but **no comparison between SPSE and RWSE expressivity is proven**. Because both aggregate over paths/walks **without enumerating them**, the Michel et al. (2023) Thm 3.3 proof strategy does not apply. So the paper’s theory establishes **RWSE has a specific blind spot** (Prop. 1) and **SPSE sees cycles** (Prop. 3) — not a clean separation theorem. The empirical table carries more weight than the theory here.

7 · Hooks into our work

💡 Thesis hook

1. The spd_bias upgrade path is now concrete. Our `GlobalAttnConv` already has the plumbing — an encoding indexed by node pair, added pre-softmax. Moving from SPD buckets to path counts is a change of **what fills the bucket**, not of architecture. The cheap intermediate is RWSE (P_k has a closed form, no counting algorithm needed); SPSE is the expensive ceiling. Worth trying RWSE first precisely because Prop. 1 tells us exactly what we’d still be missing.

💡 Thesis hook

2. Cycles are the WL blind spot too. Our `iso_wl` experiments sit at the 1-WL boundary, and WL-indistinguishable graph pairs classically differ in **cycle** structure (the standard counterexamples are two triangles vs. a 6-cycle). SPSE’s per-edge cycle spectrum is aimed straight at that. This links the message-passing ceiling (`reports/message-passing.typ`) to a concrete encoding that breaks it — and note the mechanism is **not** more depth, it is **more informative tokens**.

📝 My note

3. But it probably does not help connectedness_hard. Be honest about scope. SPSE encodes **local cyclic** structure; our hard connectivity task is about **global reach** across a bridge, and its two classes are built to be locally identical. A cycle spectrum on a dense blob’s edges says nothing about whether a bridge exists. The Sanford/Ye story (depth $\leftrightarrow$ diameter, data lever) governs that task; SPSE governs the **isomorphism / molecular-structure** side. Different axis — do not conflate them.

? Open question

4. Cost sanity check before adopting. 80 h of preprocessing for CIFAR10-scale data is real. Our graphs are small (n in the tens), so $O(KRD_{\text{DFS}}N|V|^3)$ should be minutes — but path counts must be **cached to disk alongside the dataset** (data/ is already gitignored and regenerated, so the cache slot exists). Check whether counting is even necessary at our sizes: for $n \approx 32$, **exact** enumeration at small K may be affordable and sidesteps the whole approximation-quality question.

8 · One-paragraph summary

RWSE has a provable structural blind spot: even cycles look like paths under random-walk probabilities. Replacing walk probabilities with **simple-path counts** fixes exactly that blind spot, and for adjacent pairs the counts are literally cycle counts. The price is that counting simple paths is intractable, paid down by an approximate DAG-decomposition counter that yields **lower bounds**, which works well on sparse molecular graphs and poorly on dense ones. As a drop-in with unchanged parameter count, it wins 21/24 benchmark cases. The theory does not prove SPSE $\succ$ RWSE in general — it proves one concrete gap and demonstrates the rest empirically.

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